Scalable Hybrid Quantum-Classical Optimization for the Traveling Salesman Problem

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Abstract

We will propose QUOTA, a lightweight quantum algorithm designed to obtain optimal solutions for small-scale TSP instances, and SQUARE, a hybrid quantum-classical framework that scales TSP via a divide-and-conquer strategy. Working together, they efficiently solve subproblems while extending the capability of current NISQ devices to handle larger-scale instances.

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Intro. The Traveling Salesman Problem (TSP) is a fundamental NP-hard problem with broad applications in graph databases, including path queries and route optimization [2]. Its combinatorial complexity makes it a natural target for quantum acceleration [1]. However, existing quantum approaches are still constrained by the limited qubits and circuit depth of Noisy Intermediate-Scale Quantum (NISQ) devices. This restricts them to small-scale problem instances. In this work, we propose QUOTA, a lightweight quantum algorithm that obtains optimal solutions for small-scale TSP instances with reduced quantum resource requirements. To address the scalability limitation imposed by NISQ devices, we further introduce SQUARE, a hybrid quantum-classical framework for scalable TSP solving under practical quantum constraints.

QUOTA. We propose QUOTA (QUantum Optimal paTh searching Algorithm), a lightweight algorithm for exactly solving small-scale instances of TSP on NISQ devices. The core idea behind QUOTA is to encode candidate Hamiltonian paths into a compact quantum representation, enabling efficient exploration of the solution space while minimizing qubit usage. To ensure computational accuracy, we design a dedicated mapping scheme that preserves intermediate cost information without introducing rounding errors in quantum states. Based on this representation, we construct a problem-specific oracle that filters valid Hamiltonian paths and guides amplitude amplification toward optimal solutions. Furthermore, we provide theoretical guarantees to ensure that only feasible solutions are amplified, while reducing circuit depth. These designs allow QUOTA to achieve exact solutions with high accuracy in practice, while remaining compatible with the limited resources of current devices.

SQUARE. While QUOTA is capable of obtaining exact solutions for small-scale TSP instances, its applicability is still limited by the resource constraints of NISQ devices. To address the scalability challenge, we propose SQUARE (Scalable QUantum AppRoximation framEwork), a hybrid quantum-classical framework designed to extend TSP-solving capabilities to larger instances. The core idea of SQUARE is to decompose a large problem into a set of structured subproblems that can be solved independently using QUOTA, and to coordinate their solutions through a unified framework. To

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support this process, SQUARE introduces modular quantum components for graph partitioning and subproblem planning, which ensure that subproblems remain tractable while preserving global solution quality. Specifically, we show that SQUARE achieves a bounded approximation ratio (e.g., 1.71 in typical settings), providing theoretical guarantees on solution quality. This modular design enables the flexible integration of quantum and classical routines, allowing SQUARE to adapt to different resource budgets and problem scales. As a result, SQUARE provides a practical pathway to scaling quantum TSP solving under realistic hardware constraints.

What does this yield? QUOTA and SQUARE show that combining lightweight quantum optimization with structured quantum-classical decomposition provides a practical pathway toward scaling combinatorial optimization on NISQ devices, highlighting hybrid frameworks as a promising direction to bridge the gap between current quantum capabilities and large-scale real-world applications.

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